

Supplementary Information: Extreme heat exposure around documented high-rise building fires worldwide

Author names to be confirmed

Supplementary Note 1: Evidence population and estimand

The underlying population is the set of high-rise building fire events that could be identified and documented from the searched public sources, not all high-rise fires worldwide. The primary occurrence estimand is the within-location odds ratio comparing locally extreme-hot event days with matched non-event days in the same month and weekday. It is identifiable without a global fire-reporting denominator only if, within each short matched stratum, documentation is not itself caused by day-to-day weather after conditioning on the event. Cross-country or annual raw record counts do not satisfy that condition and are therefore descriptive.

Supplementary Note 2: Cohort construction

Supplementary Table 1: **Analytical cohorts and operational definitions.**

Cohort	Size	Operational definition
All records	239	Every row in the updated workbook; no event deletion.
Valid geocode	233	Latitude in $[-90, 90]$ and longitude in $[-180, 180]$.
Extended	228	Valid date and geocode; excludes aviation, warfare, missile and comparable external triggers.
Core	191	Extended cohort with traceable official, institutional or news source and no exact duplicate candidate.

Supplementary Note 3: Exposure specification

The 1991–2020 ERA5-Land baseline is fixed before outcome analysis. The primary threshold is $\max(P_{90,local}, 30^\circ\text{C})$. A day is exposed if daily maximum temperature exceeds this threshold on that day and the two preceding days. The GEE table also retains the current day and three lags together with local P85, P90 and P95 values, so every implemented threshold and duration sensitivity is reconstructed from the same ordered daily series rather than from separate post-hoc downloads. NASA POWER sensitivity data are processed by the identical baseline, threshold and lag algorithm in a separate table; they are never combined with ERA5-Land values. The descriptive

seven-day score is

$$S = \min \left(100, 10 \sum_{d=-6}^0 \max [T_{\max,d} - \max(P_{90,local}, 30), 0] \right).$$

This bounded score was designed only for consistent map symbology. Its truncation improves visual comparability but discards variation above 10 cumulative degree-days; inferential models therefore use the binary sequence indicator and untruncated temperature anomaly.

Supplementary Note 4: Prespecified sensitivity analyses

Supplementary Table 2: **Prespecified and planned sensitivity analyses.**

Dimension	Alternative	Purpose
Heat threshold	P85, P90 and P95 with the 30 °C floor	Test dependence on relative-extreme definition.
Duration	2, 3 and 4 consecutive days at P90	Test temporal persistence assumption.
Absolute floor	30 °C versus no floor	Separate absolute thermal load from local anomaly.
Climatology	Calendar month; a 15-day calendar window is future sensitivity work	Prevent an unimplemented alternative from being presented as completed.
Lag	Current day plus three ordered lags retained in the extraction table	Test acute timing without selecting a post-hoc best curve.
Sources	Core versus extended versus all geocoded	Assess dependence on provenance criteria.
Period	Exclude 2025–2026	Assess active database-expansion period.
Event type	Exclude construction, façade or arson events in turn	Assess heterogeneous ignition pathways.
Geography	Leave-one-continent-out	Identify whether one region drives the estimate.
Weather product	NASA POWER MERRA-2/GEOS-IT	Test sensitivity to an independent, coarser reanalysis and processing stream.

Supplementary Note 5: External data provenance

ERA5-Land weather is resolved at approximately 9–11 km and should not be described as building-scale temperature. All 995 matched dates in 228 strata completed in Earth Engine. Values were sampled at the event-coordinate ERA5-Land pixel; where the land mask returned no value at a coastal point, all dates in that stratum used a documented 20-km spatial mean. Five strata required this fallback and were excluded together in sensitivity analysis. NASA POWER Release

10 meteorology is coarser, at 0.5° latitude by 0.625° longitude, and the Daily API defaults to local solar time. Its MERRA-2 stream is followed by near-real-time GEOS-IT values; 68 matched-day rows within two months of access are therefore flagged as provisional. All 1,004 matched dates returned a completed request status, 1,001 contained the complete covariate set, and the request manifest contains all 159 unique coordinates. GPW population count is used at its native 30 arc-second grid, yielding an approximately 1-km cell whose surface area varies slightly with latitude. GHSL 100-m pixels are aggregated to an approximately 1-km² neighbourhood; these represent mapped built surface, volume and average height, not the attributes of the incident building. World Bank indicators are country-year context variables with at most three years of backward matching and no forward imputation. Taiwan events remain without WDI matches.

Supplementary Table 3: **External and author-compiled data sources.**

Dataset	Support	Analytical role	Identifier and access boundary
Author-compiled events	Point/date, 2000–2026	Event outcome and reported consequence	Input SHA-256 recorded; source-specific redistribution review required.
ERA5-Land Daily Aggregated	Daily, approximately 11.1 km	Maximum temperature, dewpoint, precipitation and wind	ECMWF/ERA5_LAND/DAILY_AGGR; Copernicus acknowledgement required.
NASA POWER Release 10	Daily local solar time, 0.5° by 0.625°	Independent MERRA-2/GEOS-IT weather-product sensitivity	POWER Daily API v2.9.7; service version, source stream, request URL and access time retained.
GPWv4.11	2020, 30 arc-seconds	Native-cell population count and density	DOI: 10.7927/H4JW8BX5; CC BY 4.0.
GHSL P2023A	2018/2020, 100 m to 1 km	Built surface, volume, height and urbanisation	JRC/GHSL/P2023A; European Commission reuse notice and dataset citation apply.
World Development Indicators	Country-year	GDP, urbanisation, electricity access and national population	World Bank Indicators API v2; CC BY 4.0.
CTIF World Fire Statistics Report No. 30	Most recent national report during 2010–2023	Cross-sectional fire-service capacity sensitivity	Table 1.13; 65 countries; report page and file checksum retained; CTIF terms apply.

Supplementary Note 6: Missingness and measurement error

Coordinates are missing or invalid for six records. Injuries are numeric in 164 records and deaths in 234. The evacuation field is non-empty in 46 records but yields an exact numeric count in 43; qualitative entries such as “dozens” remain missing, and an unavailable value is not treated as

zero. Explicit unit parsing yielded metric building heights for 84 records and storey counts for 154; area-only and unlabelled numbers were not interpreted as height. In the 221-record extended-cohort observation-process sample with country-year context, grade-C sources had lower adjusted odds than grade-A sources of reporting injuries (odds ratio 0.30, 95% confidence interval 0.10–0.88) or evacuations (0.12, 0.02–0.78). Injury reporting increased with log GDP per capita (2.58 per interquartile-range increment, 1.48–4.49), whereas evacuation reporting increased with event year (4.46 per 10 years, 1.91–10.43). These exploratory models diagnose structured missingness and do not justify imputation. Descriptive event-level heat metrics combine 148 exact standardized date–coordinate matches to the prior extraction with 85 newly processed GEE records; the inferential case-control weather table was processed in the current run for all 228 strata. All failures and spatial fallbacks are retained with status codes or radius fields to prevent successful API calls from silently defining the sample.

Supplementary Table 4: **Observation-process estimates.**

Reported field	Predictor	Odds ratio	95% confidence interval	<i>P</i> value
Injuries	Grade C versus A source	0.30	0.10–0.88	0.028
Injuries	Log GDP per capita, per IQR	2.58	1.48–4.49	0.00084
Evacuations	Grade C versus A source	0.12	0.02–0.78	0.026
Evacuations	Event year, per 10 years	4.46	1.91–10.43	0.00055

Both binomial generalized linear models used 221 observations and adjusted simultaneously for event year, log GDP per capita, source grade and continent with heteroskedasticity-consistent standard errors. Numeric injury and evacuation counts were available for 158 and 42 modelled records, respectively. The reference categories were grade-A sources and Asia. IQR denotes interquartile range.

Supplementary Note 7: Model diagnostics

The ERA5-Land conditional logistic models converged for all six prespecified binary heat definitions using 995 matched days in 228 strata. Only 12–46 strata were informative for a given definition, explaining the wide confidence intervals; all intervals included one (Supplementary Table 5). Excluding all five strata that required the 20-km coastal fallback yielded an odds ratio of 0.77 (95% confidence interval 0.25–2.36; $P = 0.651$), compared with 0.72 (0.24–2.18; $P = 0.563$) in the complete primary sample. All 12 ERA5-Land cohort, event-type and leave-one-continent-out fits converged and included one. The NASA POWER models likewise converged for all six definitions. Their adjusted sample contained 1,001 matched days in 228 strata; the three omitted rows lacked at least one time-varying covariate, and only 7–43 strata were informative. No definition from either product crossed a conventional significance threshold. Alternatives are treated as a prespecified family rather than searched for a preferred estimate. A model is not selected solely because it maximizes in-sample R^2 ; uncertainty and product disagreement are retained.

Supplementary Table 5: **ERA5-Land model results.**

Heat definition	Odds ratio	95% confidence interval	Informative strata	<i>P</i> value
P90, 3 days + 30 °C floor	0.72	0.24–2.18	21	0.563
P85, 3 days + 30 °C floor	0.87	0.34–2.20	27	0.769
P90, 2 days + 30 °C floor	1.21	0.54–2.70	30	0.641
P90, 4 days + 30 °C floor	0.93	0.26–3.35	14	0.912
P95, 3 days + 30 °C floor	0.33	0.04–2.57	12	0.290
P90, 3 days, no floor	0.96	0.47–1.96	46	0.918

All models use 995 matched days in 228 strata and adjust for dewpoint, precipitation and 10-m wind speed, each scaled per interquartile range. Odds ratios compare an exposed with an unexposed matched day. Confidence intervals are Wald intervals and *P* values are two-sided.

Supplementary Table 6: **NASA POWER model results.**

Heat definition	Odds ratio	95% confidence interval	Informative strata	<i>P</i> value
P90, 3 days + 30 °C floor	1.14	0.39–3.36	17	0.807
P85, 3 days + 30 °C floor	1.09	0.44–2.68	25	0.849
P90, 2 days + 30 °C floor	1.22	0.52–2.90	26	0.648
P90, 4 days + 30 °C floor	1.37	0.39–4.77	12	0.622
P95, 3 days + 30 °C floor	1.32	0.25–6.84	7	0.741
P90, 3 days, no floor	1.35	0.68–2.67	43	0.394

All models adjusted for dewpoint, precipitation and 10-m wind speed, each scaled per interquartile range. Odds ratios compare an exposed with an unexposed matched day. Confidence intervals are Wald intervals and *P* values are two-sided. No multiplicity-adjusted inferential claim is made for the alternatives.

Supplementary Note 8: Recorded consequence heterogeneity

The concentration analysis is restricted to numeric values in the extended cohort and therefore describes the distribution of reported burden, not population risk. The 225 numeric death records sum to 1,026 deaths and have a Gini coefficient of 0.889; the 162 numeric injury records sum to 2,031 injuries and have a Gini coefficient of 0.766. Reported building size is a relative evidence measure formed from the mean within-dataset percentile rank of explicit height and explicit storey count, using the available rank when only one measure is present. It avoids treating area-only or unlabelled values as height but does not recover missing physical dimensions.

The adjusted death model contains 177 records, of which 74 report a positive count. Because 98.7% of the extended cohort has a numeric death value, the model uses equal weights. The injury model contains 127 records, of which 85 report a positive count, and uses stabilized inverse-probability weights from the injury-reporting model; the post-truncation weights range from 0.80 to 1.80 and give an effective sample size of 120.6. Both models use $\log(\text{count}+1)$, adjust simultaneously for reported building size, event year, log GDP per capita, construction, façade, arson, source grade

and continent, and use HC3 standard errors. Exponentiated coefficients are ratios of the geometric mean of recorded outcome plus one. Weighting can address selection only under the untestable assumption that measured reporting covariates are sufficient; it does not remove catalogue selection, outcome misclassification or missing building-size bias.

A separate GEE-linked severity model required both deaths and injuries to be numerically reported before summing them. Sixty-six core-cohort events also had complete heat score, population, built volume and event year. The negative-binomial model used this parsimonious adjustment set because façade and construction indicators occurred in only two and five complete cases, respectively, and forcing them together with multi-level factors produced unstable separation-like estimates. The heat-score rate ratio was 1.04 per 10 points (95% confidence interval 0.90–1.21; $P = 0.608$; Supplementary Table 7). This event-level result is exploratory because the bounded score was designed for mapping, complete-case selection is substantial and the outcome combines heterogeneous reported consequences.

Supplementary Table 7: **Exploratory heat-score and reported total-casualty model.**

Predictor and scale	Rate ratio	95% confidence interval	P value
Heat score, per 10 points	1.04	0.90–1.21	0.608
Population count, per log unit	1.33	1.01–1.75	0.040
Built volume, per log unit	0.28	0.16–0.50	< 0.001
Event year, per decade	2.77	0.18–43.24	0.467

The model uses a negative-binomial mean structure with HC3 standard errors. Ratios are conditional associations among 66 complete documented events, not incidence or causal effects; no multiplicity-adjusted claim is made.

Supplementary Table 8: **Recorded consequence models.**

Outcome	Predictor and scale	Ratio	95% confidence interval	P value
Deaths	Reported building size, per IQR	0.81	0.59–1.11	0.197
Deaths	Event year, per 10 years	0.91	0.69–1.19	0.477
Deaths	Log GDP per capita, per IQR	0.60	0.38–0.95	0.028
Deaths	Construction-related	0.58	0.34–1.00	0.048
Deaths	Façade fire	0.60	0.36–1.00	0.050
Deaths	Arson trigger	1.19	0.49–2.89	0.697
Injuries	Reported building size, per IQR	0.74	0.44–1.25	0.261
Injuries	Event year, per 10 years	0.53	0.32–0.87	0.013
Injuries	Log GDP per capita, per IQR	1.01	0.52–1.98	0.967
Injuries	Construction-related	0.31	0.08–1.21	0.092
Injuries	Façade fire	0.25	0.11–0.57	0.0010
Injuries	Arson trigger	1.12	0.33–3.76	0.859

The ratios compare the geometric mean of recorded outcome plus one. Source-grade and conti-

nent coefficients are included in the fitted models but omitted from the table to keep attention on prespecified substantive factors. All P values are two-sided and unadjusted; the entire analysis is exploratory.

Supplementary Note 9: Fire-service capacity sensitivity

CTIF Table 1.13 was parsed as 65 continuous country rows and linked by ISO3 code. Career- and total-firefighter metrics covered 141 of 228 extended-cohort events across 26 countries; fire-station and fire-engine metrics covered 111 events in 24 countries and 119 events in 24 countries, respectively. Coverage was geographically asymmetric, including no matched extended-cohort event in Africa or South America. The table reports each country’s most recent available value during 2010–2023, so the metrics neither share a common reference year nor measure resources deployed to the focal incident.

Each sensitivity model added one log-transformed capacity metric, scaled per interquartile range, to the recorded-consequence specification in Supplementary Note 8. All eight 95% confidence intervals included one (Supplementary Table 9; Supplementary Fig. 4). This imprecision, together with partial coverage and country-level exposure assignment, precludes a claim of either benefit or absence of effect. The analysis instead quantifies what the available cross-national resource data can and cannot support.

Supplementary Table 9: **National fire-service capacity sensitivity models.**

Outcome	Capacity metric, per IQR of log(count per 100,000 + 1)	Ratio	95% confidence interval	Events	Countries
Deaths	Career firefighters	0.93	0.62–1.39	120	22
Deaths	All firefighters	0.98	0.82–1.16	120	22
Deaths	Fire stations	0.64	0.39–1.04	93	20
Deaths	Fire engines	0.73	0.45–1.20	99	20
Injuries	Career firefighters	1.26	0.56–2.82	86	19
Injuries	All firefighters	0.90	0.70–1.15	86	19
Injuries	Fire stations	0.95	0.42–2.13	66	17
Injuries	Fire engines	1.17	0.49–2.81	71	17

Ratios compare the geometric mean of recorded outcome plus one and are adjusted for reported building size, event year, log GDP per capita, construction, façade, arson, source grade and continent. Injury models use stabilized numeric-reporting weights; death models use equal weights. All intervals use HC3 standard errors, and all tests are exploratory and unadjusted for multiplicity.

Supplementary Note 10: Evidence-window and geographic sensitivity

The ERA5-Land primary heatwave definition was refitted under six evidence and event restrictions and six leave-one-continent-out restrictions. All 12 models converged. Cohort and event-restriction

estimates ranged from 0.23 after excluding 2025–2026 to 0.92 in the core cohort; leave-one-continent-out estimates ranged from 0.41 after excluding Asia to 0.86 after excluding Africa. Every 95% confidence interval included one, with 9–21 informative strata (Supplementary Table 10). The independent NASA POWER analysis was refitted under the same restrictions. Its estimates ranged from 0.94 to 1.81 for cohort or event exclusions and from 0.76 to 1.56 for leave-one-continent-out analyses; all intervals again included one, with 9–17 informative strata (Supplementary Table 11; Supplementary Fig. 5). The two products differ in point-estimate direction, but agree that the available matched sets provide low information for sustained extreme heat.

Supplementary Table 10: **ERA5-Land evidence-window and geographic sensitivity.**

Restriction	Odds ratio	95% confidence interval	Informative strata
Extended cohort	0.72	0.24–2.18	21
Core cohort	0.92	0.29–2.89	17
Exclude 2025–2026	0.23	0.03–1.78	13
Exclude construction	0.86	0.28–2.66	18
Exclude façade	0.88	0.25–3.14	15
Exclude arson	0.76	0.25–2.32	20
Leave out Asia	0.41	0.05–3.31	9
Leave out Europe	0.80	0.26–2.42	20
Leave out Oceania	0.72	0.24–2.17	21
Leave out North America	0.63	0.18–2.23	17
Leave out Africa	0.86	0.28–2.68	18
Leave out South America	0.77	0.25–2.33	20

All models use the P90, three-day definition with a 30 °C floor and adjust for dewpoint, precipitation and wind. A separate complete exclusion of the five 20-km coastal-fallback strata gave an odds ratio of 0.77 (0.25–2.36) from 20 informative strata.

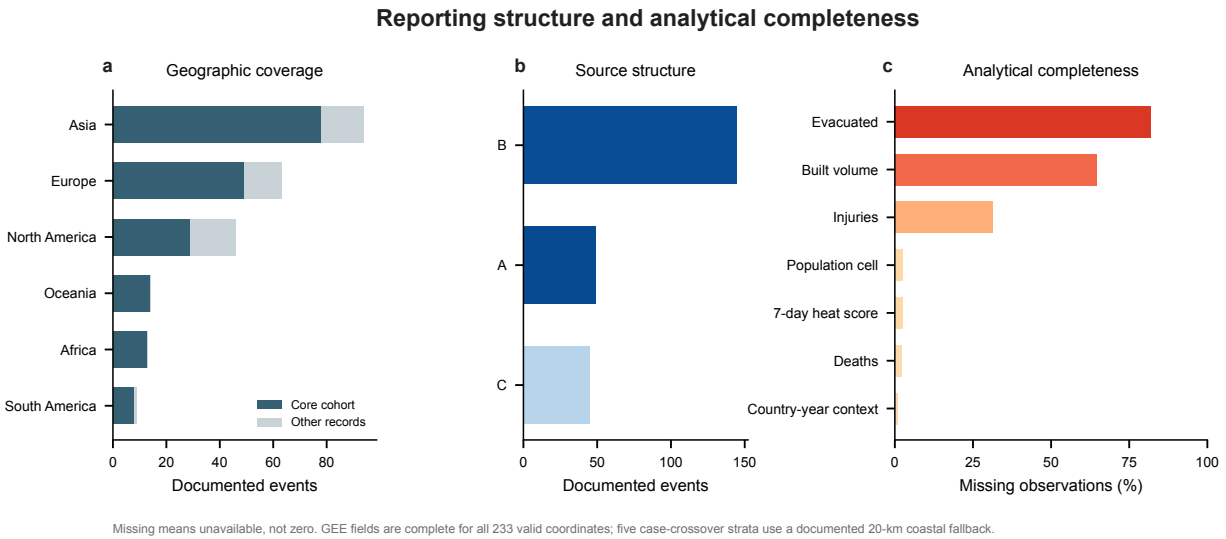
Supplementary Table 11: **Evidence-window and geographic sensitivity.**

Restriction	Odds ratio	95% confidence interval	Informative strata
Extended cohort	1.14	0.39–3.36	17
Core cohort	0.94	0.29–3.09	15
Exclude 2025–2026	1.14	0.30–4.30	12
Exclude construction	1.36	0.45–4.14	15
Exclude façade	1.81	0.53–6.20	11
Exclude arson	1.13	0.38–3.32	17
Leave out Asia	0.84	0.17–4.11	9
Leave out Europe	1.56	0.50–4.82	14
Leave out Oceania	1.17	0.40–3.45	17

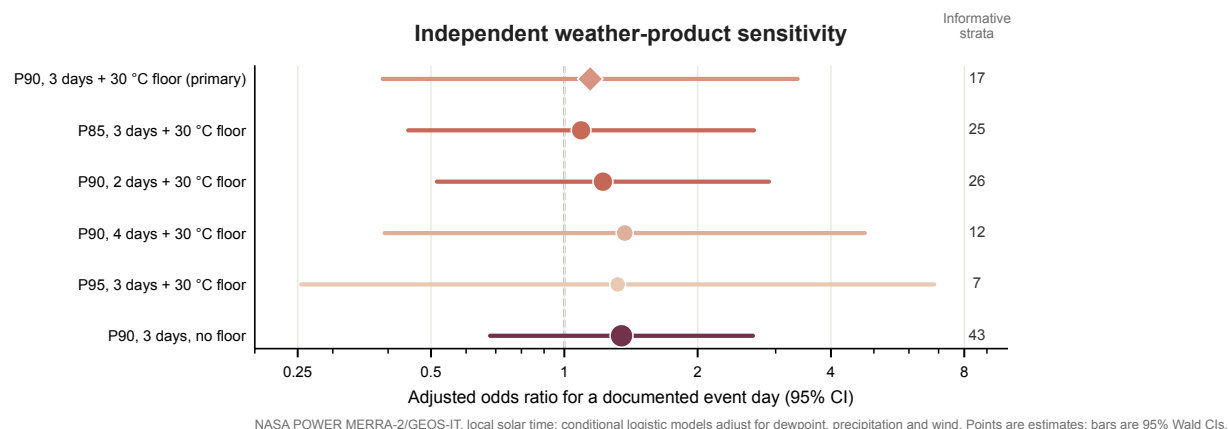
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Restriction	Odds ratio	95% confidence interval	Informative strata
Leave out North America	1.15	0.39–3.38	17
Leave out Africa	0.76	0.20–2.93	13
Leave out South America	1.37	0.45–4.17	15

All models use the P90, three-day definition with a 30 °C floor and adjust for dewpoint, precipitation and wind. Odds ratios compare a heatwave-exposed with an unexposed matched day. Confidence intervals are Wald intervals; no multiplicity-adjusted inference is made.

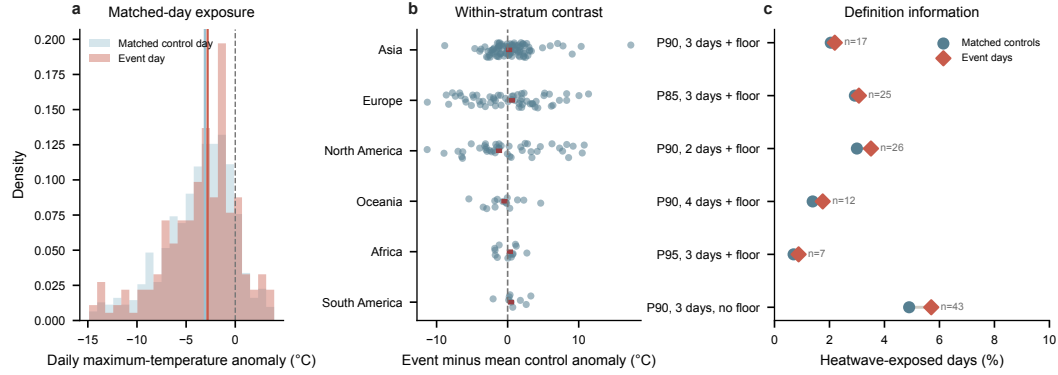


Supplementary Figure 1: **Reporting structure and analytical completeness.** **a**, Documented and core-cohort events by continent. **b**, Records by operational source grade; colour depth represents the core-cohort share. **c**, Missingness of reported consequence variables, country-year context and Earth Engine-linked variables after authenticated processing of all valid coordinates. Missing casualty values mean unavailable information and are not recoded as zero. Panel-level source data are exported by the figure notebook.



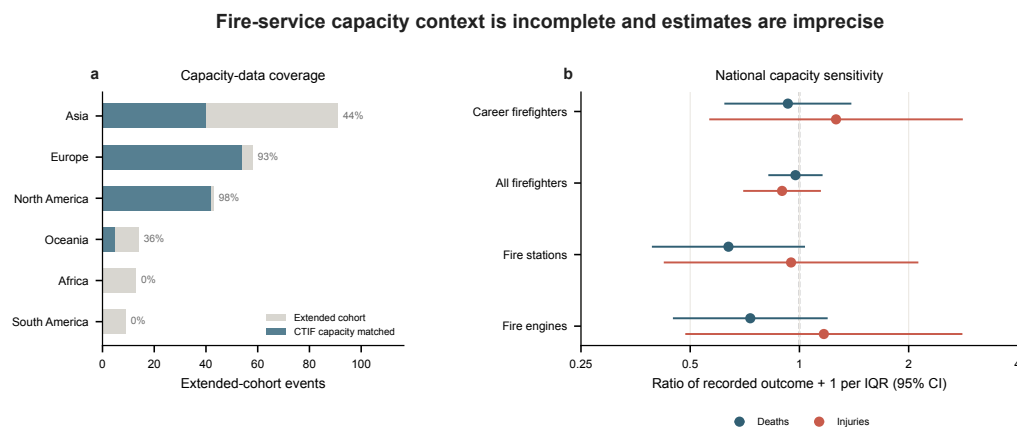
Supplementary Figure 2: **Independent weather-product sensitivity.** Adjusted odds ratios compare a heatwave-exposed with an unexposed documented event day in time-stratified conditional logistic models using NASA POWER MERRA-2/GEOS-IT meteorology. Points show estimates, horizontal bars show 95% Wald confidence intervals, colour depth and point area encode the number of informative strata, and the diamond marks the prespecified primary definition. All models use 1,001 matched days in 228 strata and adjust for dewpoint, precipitation and wind; informative strata vary by heat definition as shown. Source data are provided in the panel-level source-data file.

Matched-weather diagnostics for the independent product



NASA POWER MERRA-2/GEOS-IT in local solar time. Controls are same-weekday dates in the same month and location; panel c weights each stratum equally. Diagnostics are descriptive, whereas inference uses conditional logistic regression.

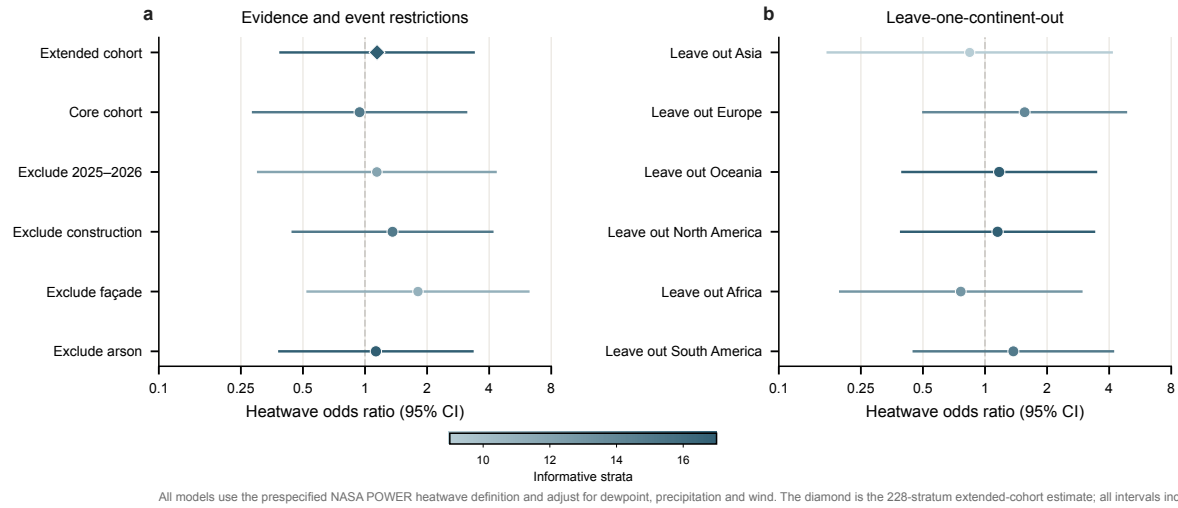
Supplementary Figure 3: **Matched-weather diagnostics for the independent product.** **a**, Density distributions of daily maximum-temperature anomalies on event days and matched control days; vertical lines mark group medians. **b**, Event-day anomaly minus the mean anomaly of matched control days in each stratum, grouped by continent; points are strata and red squares mark continent medians. **c**, Equal-stratum-weighted percentage of event days and matched control days classified as heatwave-exposed under the six prespecified definitions; labels give the number of informative strata. NASA POWER MERRA-2/GEOS-IT data use local solar time, and controls share the event location, calendar month, year and weekday. These panels diagnose exposure structure; inferential estimates come from the conditional logistic models in Supplementary Table 6.



CTIF Report No. 30 Table 1.13 provides each country's most recent reported value during 2010–2023, not a common-year series. Models add one log-scaled capacity metric at a time to the recorded-consequence specification; all intervals include one.

Supplementary Figure 4: **Coverage and sensitivity of national fire-service capacity data.** **a**, Extended-cohort event counts and the subset matched to a CTIF career-firefighter metric by continent; percentages denote event-level coverage. **b**, Exploratory adjusted ratios of recorded outcome plus one after adding one national capacity metric at a time. Points show estimates and horizontal bars show 95% heteroskedasticity-consistent confidence intervals. CTIF Report No. 30 Table 1.13 supplies each country's most recent value during 2010–2023, not a common-year series. All intervals include one, and the estimates must not be interpreted causally or as evidence of no effect.

Coarse-product heat estimates remain imprecise across restrictions



Supplementary Figure 5: **Evidence-window and geographic sensitivity of the independent weather product.** **a**, The NASA POWER primary heatwave definition in the extended and core cohorts and after prespecified period or event-type exclusions. **b**, The same model after leaving out each continent in turn. Points show adjusted odds ratios, horizontal bars show 95% Wald confidence intervals, and colour and area encode the number of informative strata. The diamond marks the extended-cohort estimate. Every model adjusts for dewpoint, precipitation and wind; all intervals include one.